

Syntactic Forcing Models for Coherent Logic

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A category of forcing conditions

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Categories of forcing conditions

- ▶ Fix a finite first-order signature Σ
- ▶ Fix a countably infinite set of variables $X = \{x_0, x_1, \dots\}$
- ▶ Let $\text{Tm}(X)$ be the set of Σ -terms over $X \subseteq X$
- ▶ Define the category $\mathbb{C}_{\text{ts}}$ having:
 - ▶ Objects denoted as pairs $(X; A)$, where X is a finite subset of X and A is a finite set of atoms in the language defined by Σ and X . (This means that only variables from X may occur in A .) Such pairs $(X; A)$ are called *conditions*.
 - ▶ Morphisms denoted as $f : (Y; B) \rightarrow (X; A)$, where f is a *term substitution* $X \rightarrow \text{Tm}(Y)$ such that $Af \subseteq B$
 - ▶ Composition $f \circ g$ of $g : (Z; C) \rightarrow (Y; B)$ with f above is the substitution $X \rightarrow \text{Tm}(Z)$ that is the composition fg (in diagram order!) of the respective substitutions
 - ▶ Identity morphisms $(X; A) \rightarrow (X; A)$ are identity substitutions $X \rightarrow X$
- ▶ Similarly to $\mathbb{C}_{\text{ts}}$, define $\mathbb{C}_{\text{vs}}$ ($\mathbb{C}_{\text{m}}$) when in addition $f(X) \subseteq Y$ (and also f injective)

Categories of forcing conditions (ctnd)

- ▶ $\mathbb{C}_{ts}, \mathbb{C}_{vs}, \mathbb{C}_{rn}$ are indeed categories with terminal object $(;)$
- ▶ Conditions will be denoted as, e.g., $(x, y, z; p(z), q(f(x), z, z))$
- ▶ Substitutions will be denoted as, e.g., $[y := x, z := g(x)]$
- ▶ Post-fixing substitutions in diagram order: $(Af)g = A(fg)$
- ▶ Depending on Σ , categories $\mathbb{C}_{ts}, \mathbb{C}_{vs}, \mathbb{C}_{rn}$ are rather different:
 - ▶ $[x := 0], [x := 1] : (;) \Rightarrow (x;)$ cannot be equalized
 - ▶ $[x := y], [x := z] : (y, z;) \Rightarrow (x;)$ can be equalized by $[y := w, z := w] : (w;) \rightarrow (y, z;)$ in $\mathbb{C}_{vs}$, but not in $\mathbb{C}_{rn}$
- ▶ Actually, $\mathbb{C}_{ts} (\mathbb{C}_{vs})$ has all finite products (limits)
- ▶ Depending on Σ , categories $\mathbb{C}_{ts}, \mathbb{C}_{vs}, \mathbb{C}_{rn}$ will lead to different forcing semantics (good for independence proofs!)
- ▶ What does a condition mean? A finite, partial description of potential models. Time to consider a coherent theory T ...

Coverages depending on coherent theories

- Fix a coherent theory T
- Define inductively a relation $\triangleleft_T$ between conditions and finite sets of conditions (denoted by $U, V, \dots$):
 - (base) $C \triangleleft_T \{C\}$ for all conditions C
 - (step) If T has an axiom $\forall \vec{x}. (C \rightarrow (\exists \vec{y}_1. B_1) \vee \dots \vee (\exists \vec{y}_n. B_n))$ such that for some sequence of terms $\vec{t}$ with variables in X we have $C[\vec{t}/\vec{x}] \subseteq A$, then the following rule applies:

$$\frac{(X, \vec{y}_1; A, B_i[\vec{t}/\vec{x}]) \triangleleft_T U_1 \quad \dots \quad (X, \vec{y}_n; A, B_i[\vec{t}/\vec{x}]) \triangleleft_T U_n}{(X; A) \triangleleft_T \bigcup_{1 \leq i \leq n} U_i}$$

- Looks familiar? Let's take the semantic point of view.
- Example: if $T = \{p \rightarrow (q \vee r)\}$, then $(; p) \triangleleft_T \{(; p, q), (; p, r)\}$. The models of T extending $(; p)$ are models extending $(; p, q)$ or models extending $(; p, r)$
- Borderline case: if $T = \{p \rightarrow \perp\}$, then $(; p) \triangleleft_T \emptyset$
- When $C \triangleleft U$ (drop $_T$, also: $U \triangleright C$) we say that U covers C

Structural properties of the coverage

- ▶ The properties of $\triangleleft$ use (any one of) $\mathbb{C}_{ts}, \mathbb{C}_{vs}, \mathbb{C}_m$
- ▶ Lemma $\triangleleft 1$. If $(X; A) \triangleleft U$ and $(Y; B) \in U$, then $X \subseteq Y$ and $A \subseteq B$ and $i_{X,Y} : (Y; B) \rightarrow (X; A)$. Prf: easy induction on $\triangleleft$.
- ▶ Lemma $\triangleleft 2$. If $f : D \rightarrow C$ and $C \triangleleft U$, then there is $V \triangleright D$ such that, for any $E \in V$ there is an $F \in U$ such that $g : E \rightarrow F$ with g an extension of f . Proof: induction on $\triangleleft$. Intuition: view D as an (extension of) the f -instance of C .
NB $\mathbb{C}_m$ OK!
- ▶ Lemma $\triangleleft 3$. If $C \triangleleft U$ and for every $D \in U$ we have a $V_D \triangleright D$, then $C \triangleleft \bigcup_{D \in U} V_D$. Proof: induction on $\triangleleft$. Intuition: transitivity.
- ▶ Together with $\triangleleft 0 : C \triangleleft \{C\}$, $\triangleleft 0$ – $\triangleleft 3$ provide what is needed for the coming definition of forcing to give a sound and complete semantics.
- ▶ Further abstraction $\rightsquigarrow$ Grothendieck topology and site

Forcing relation based on coverage

Let $\triangleleft$ be a coverage. For any condition $C = (X; A)$ and any first-order formula ϕ with free variables in X , we define the forcing relation $C \Vdash \phi$ by induction on ϕ as follows:

1. $C \Vdash \top$
2. $C \Vdash \perp$ if $C \triangleleft \emptyset$ (i.e., $A \vdash_X \perp$, explain)
3. $C \Vdash \phi$ if ϕ is an atom and there is $U \triangleright C$ such that $\phi \in B$ for all $(Y; B) \in U$ (i.e., $A \vdash_X \phi$)
4. $C \Vdash \phi_1 \wedge \phi_2$ if $C \Vdash \phi_1$ and $C \Vdash \phi_2$
5. $C \Vdash \phi_1 \vee \phi_2$ if for some U we have $C \triangleleft U$ and ($D \Vdash \phi_1$ or $D \Vdash \phi_2$) for all $D \in U$
6. $C \Vdash \phi_1 \rightarrow \phi_2$ if for all D and morphisms $f : D \rightarrow C$ we have $D \Vdash \phi_2 f$ whenever $D \Vdash \phi_1 f$
7. $C \Vdash \forall x. \phi$ if for all $D = (Y; B)$ and morphisms $f : D \rightarrow C$ we have $D \Vdash \phi[f, x = t]$ for all $t \in \text{Tm}(Y)$
8. $C \Vdash \exists x. \phi$ if there is $U \triangleright C$ such that, for all $D \in U$, $D = (Y; B)$, $D \Vdash \phi[x = t]$ for some $t \in \text{Tm}(Y)$

Examples

- ▶ The law of the excluded middle is not forced: for $\Sigma = \{p\}$, $T = \emptyset$, **not** $(;) \Vdash p \vee \neg p$
- ▶ Unlike Kripke semantics, there is no one-world frame. Hence for $\Sigma = \{p\}$, $T = \emptyset$, surprisingly, $(;) \Vdash \neg\neg p$
- ▶ Classical contingencies can sometimes be forced: for $\Sigma = \{P(-)\}$, $T = \emptyset$, **never** $C \Vdash \forall x.P(x)$, so $(;) \Vdash (\forall x.P(x)) \rightarrow \perp$
- ▶ Distinguishing $\Vdash_{\text{vs}}$ and $\Vdash_{\text{ts}}$: if $\Sigma = \{Z(-), 0\}$ and
 - ▶ $T = \{\neg Z(0)\}$, then $(x;) \Vdash_{\text{vs}} \neg\neg Z(x)$, and **not** $(x;) \Vdash_{\text{ts}} \neg\neg Z(x)$ (since $[x := 0] : (;) \rightarrow (x;)$, $(;) \Vdash_{\text{ts}} \neg Z(0)$ and **not** $(;) \Vdash_{\text{ts}} \perp$)
 - ▶ Better, add $\exists x.\top$ to T and get for $\phi = \exists x.\neg\neg Z(x)$ that $(;) \Vdash_{\text{vs}} \phi$ and **not** $(;) \Vdash_{\text{ts}} \phi$.
- ▶ NB $T \vdash \exists x.\top$ and yet it makes a difference
- ▶ Distinguishing $\Vdash_{\text{rn}}$ from $\Vdash_{\text{vs}}$, $\Vdash_{\text{ts}}$ is done in [BBC, 6.4] by a rather complicated example (with relational Σ)

Special soundness: forcing the theory itself

- ▶ Fix a coherent theory T with its $\triangleleft$ and $\Vdash$
- ▶ For all $\phi \in T$ we have $(;) \Vdash \phi$
- ▶ Proof by example: take $\phi \equiv \forall x. (P(x) \rightarrow (p \vee \exists y. Q(x, y)))$.
(TL;DR) Note that $(;)$ is final, so we have to show that $C \Vdash P(t) \rightarrow p \vee \exists y. Q(t, y)$ for all conditions $C = (X; A)$ and $t \in \text{Tm}(X)$. So, we have to show that $D \Vdash (p \vee \exists y. Q(tf, y))$ for all $D = (Y; B)$ and $f : D \rightarrow C$ with $D \Vdash P(tf)$. Now, if $U \triangleright D$ such that every $E \in U$ contains $P(tf)$, then we can use the instance of ϕ with tf to cover E such that $E \Vdash p \vee \exists y. Q(tf, y)$, and use $\triangleleft_3$ to get $D \Vdash p \vee \exists y. Q(tf, y)$.
- ▶ By the general soundness result (next slide), not only T is forced, but also all its intuitionistic, possibly non-coherent consequences.

General soundness of the forcing semantics

- ▶ Fix a signature Σ , one of the categories $\mathbb{C}_{ts}, \mathbb{C}_{vs}, \mathbb{C}_{rn}$, a coverage $\triangleleft$ with its forcing relation by $\Vdash$
- ▶ **No** coherent theory T is assumed here
- ▶ Let $\Gamma \vdash_X^i \phi$ denote intuitionistic provability (explain X)
- ▶ Soundness: for **all** formulas Γ, ϕ with free variables in X , if $\Gamma \vdash_X^i \phi$, then for any C and $\rho : X \rightarrow \text{Tm}(C)$,

$$C \Vdash \Gamma \rho \text{ implies } C \Vdash \phi \rho$$

- ▶ Proof: induction on $\Gamma \vdash_X^i \phi$ (long and tedious)

Completeness for coherent formulas

- ▶ Fix a coherent theory T with its $\triangleleft$ and $\Vdash$
- ▶ Coherent completeness: for every **coherent** sentence ϕ , if $(,) \Vdash \phi$, then $T \vdash_{\emptyset}^i \phi$
- ▶ For the proof we need a version for open formulas
- ▶ Completeness: for every **coherent** sentence ϕ with free variables in X , any condition $C = (Y; A)$ and $\rho : X \rightarrow \text{Tm}(Y)$,

$$C \Vdash \phi\rho \text{ implies } T, A \vdash_X^i \phi\rho$$

- ▶ Proof by induction on ϕ
- ▶ This proof is constructive, and doesn't use 'fairness'
- ▶ On the other hand, there is syntax in this semantics

Redundant sentences

- ▶ Let T be a coherent theory. A sentence ϕ is called *T-redundant* if all coherent sentences ψ such that $T, \phi \vdash^i \psi$ can be proved already in T
- ▶ The combination of soundness with coherent completeness yields that ϕ is redundant if ϕ is forced: if $T \vdash^i \phi \rightarrow \psi$, then by soundness $\phi \rightarrow \psi$ is forced. Hence if ϕ is forced, then also ψ is forced, and hence provable in T if coherent.
- ▶ Example (Kock): in the theory of local rings, the following formula is forced and hence redundant (surprise?)

$$\neg(x = 0 \wedge y = 0) \rightarrow (\exists z. xz = 1) \vee (\exists z. yz = 1)$$